$k_0 := (k_0-1)a < b,\ k_0 a \geq b$

$S = \{0 \leftrightarrow a \hookrightarrow 2a \cdots k_0 a \hookrightarrow k_0 a + a \cdots\}$, directed

$k_0 a - b \quad k_0 a - b + 1$

$\downarrow$

$2(k_0 a - b) \cdots$

$\downarrow$

all the states

want $x \in S$ / forbidden

$x \cup \text{forbidden} \in S$

$\max\{x \cup \text{forbidden}\} \leq \hat{c}(k_0 a - b) + \tilde{r} a$

$\max\{x \cup \text{forbidden}\} + a + b \leq (\tilde{c} k_0 + \tilde{r} + 1) a + (1 - \hat{c}) b$

can preserve $(\tilde{c} k_0 + \tilde{r} + 1) a + (1 - \hat{c}) b > 0$

$\Rightarrow$ Least upper bound